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Title of the article: *Whole-lung low-dose radiation therapy (LD-RT) for non-intubated oxygen-dependent patients with COVID-19-related pneumonia receiving dexamethasone and/or remdesivir*

The content extracted is as follows:

A recent but small randomized trial studied LD-RT in 11 intubated patients and detected no benefit compared to 11 intubated controls [20].

Materials and methods

Patients underwent clinical assessment at the time of enrollment and on post-RT days 1, 3, 7, and 28, as well as optional assessment on days 14 and 21.

Evaluation of serum inflammatory, renal, cardiac, chemistry, clotting, and hematologic markers were encouraged daily, but obtained at least at baseline and at the timepoints mentioned above. Following trial completion, a prospective cohort of 20 contemporaneous controls was retroactively and blindly matched by age, comorbidity burden, and disease severity for comparative outcome analysis.

Patients in the control cohort received best supportive care and standard of care drug therapies for COVID-19 (i.e., glucocorticosteroids, remdesivir, etc.) per physician discretion or therapeutic protocol without LD-RT.

Start time to clinical recovery was defined as it was in prior cohorts [17], as the time from LD-RT delivery (in the radiation cohort), as the first day of administration of COVID-19 therapy (in control patients, if received [$n = 18$]), or as the first full-day of hospitalization (in control patients who received best supportive care alone [$n = 2$]).

To control for lead-time bias from this definition, hospital duration and time from hospital admission to clinical recovery were also evaluated. Days febrile and oxygenation duration and burden were tracked. During trial enrollment, durable declines in C-reactive protein (CRP) were observed following LD-RT in most recovering patients, while CRP rise occurred prior to poor outcomes.

A subset analysis was planned mid-trial, defining a predictive variable CRP response. Patients who did and did not experience consistent decline in CRP over 3 days following LD-RT were categorized as rewire defined as experiencing both of the following: (1) elevated CRP level (>10 mg/dL) the morning prior to LD-RT (baseline), and (2) two sequential rises in CRP above baseline among the first three daily measurements immediately following LD-RT.

Evaluation was permitted to span 4 days following LD-RT in the event of missing labs.

Median and interquartile range was calculated for laboratory values at clustered time points: hospital days 0, 2, 3, 5, 6, 8, 9, 11, and 12–14.

Results

From June 11, 2020 to December 7, 2020, 65 patients were screened for eligibility, and twenty-five were enrolled. Of these, the first twenty were treated with LD-RT and 5 were reserved as future controls for immunologic profiling.

Following trial completion, 20 control patients hospitalized between March 6, 2020 and November 19, 2020 and enrolled on a separate collaborating non-therapeutic trial were blindly matched for comparison against the LD-RT group.

One control rapidly intubated on the day of admission and later died and was excluded and blindly replaced to ensure matching of disease severity at admission between cohorts.

Median duration of symptoms prior to admission was 10 (range 1–15) and 7 (2–14) days following LD-RT and for controls, respectively ($p = 0.10$).

Patients in both the LD-RT and control cohorts received COVID-directed drug therapy for a median of 7.5 days ($p = 0.87$).

LD-RT was administered at median hospital day 3 (range 1–8).

Patients in the LD-RT cohort received COVID-directed therapy (dexamethasone and/or remdesivir, etc.) at a median start time of hospital day 1 (range 1–5), compared to 2.5 (range 1–6) in controls ($p = 0.02$).

Intubation rates reduced by 56% following LD-RT compared to controls (14% vs.

32% in controls), translating to freedom from intubation at day 28 (Fig. 2) of 86% following LD-RT compared to 68% in controls, which approached statistical significance ($p = 0.09$).

Median hospital duration was 10.5 days (range 5–33) in the LD-RT cohort compared to 11.5 days (range 3–42) in controls ($p = 0.61$).

Additional treatment outcomes for the entire cohort are reported in Table 2. Radiographic improvement was not detectable between the LDRT and control cohorts ($p = 0.72$, Fig. 4).

Plotted medians and inter-quartile ranges of serologic biomarkers for both LD-RT and control patients are shown in Fig.

Decline of inflammatory biomarker C-reactive protein (CRP) was statistically superior in the LD-RT cohort compared with controls ($p = 0.02$).

Cardiac marker, creatine kinase, started higher in LD-RT patients, but was also significantly reduced over controls ($p < 0.01$).

Liver function remained normal following LD-RT, while transaminitis occurred in some controls, but did not reach significance (AST = 0.35; ALT $p = 0.60$).

Myoglobin, erythrocyte sedimentation rate, lactate dehydrogenase, monocyte count, ferritin, brinogen, procalcitonin, and interleukin-6 trended downward after LD-RT but did not reach significance or control comparisons were not different or not available (Fig. 6).

Sixteen of 20 irradiated patients (80%) experienced durable decline in CRP levels over 3 days following LD-RT and were classified as CRP responders.

Outcomes for these 16 LD-RT responders compared with their 16 matched controls are shown in Fig. 3 and Table 3 ($n = 32$).

Freedom from intubation was 100% for LD-RT responders compared to 66% for matched controls ($p = 0.01$, Table 3, Fig. 3d), translating to a reduction in intubation from 34% to 0%.

As a categorical variable, none of the LD-RT responders died or required intubation, compared to 5 of 16 matched controls ($p = 0.04$, Table 3).

Median duration of time intubated was 0 days for LDRT responders vs.

Median time to clinical recovery was 7.5 days (range 2–15) in LD-RT responders and 7 days (range 3–38) in controls but maximum recovery time was reduced (15 vs 38 days), shown as upper curve separation ($p = 0.29$, Fig. 3a).

Median time from admission to clinical recovery was 9.5 days (range 1–17) following LD-RT compared to 9 days (range 1–41) in controls, with reduction in maximum recovery time (17 vs.

Median hospital duration was 10 days (range 5–18) for LD-RT responders compared to 10 days (range 3–42) for controls, with reduced maximum hospitalization (18 vs 42 days) and upper curve separation ($p = 0.22$, Fig. 3c).

The proportion of patients requiring prolonged hospitalization of 3 weeks or longer fell from 30% to 10% with the addition of LD-RT ($p = 0.24$). Of the 4 patients who were CRP non-responders in the LD-RT cohort, 2 required intubation, 3 died by day 28 (1 refused intubation), and the 4th died on day 33.

Six of 18 CRP responder controls (33%) required intubation despite post-drug CRP decline and both of the non-responders recovered without intubation ($p = 0.36$). Mean total time requiring oxygen supplementation prior to recovery was 10.5 days (range 4–18) in LD-RT responders compared to 14.3 days (range 3–42) in controls ($p = 0.24$).

Average daily oxygen maximums per patient were 32% lower at 7.6 L/min (mean) in LD-RT responders compared to 24 L/min in controls ($p = 0.03$, Table 3).

3.3 days (range 0–12), respectively ($p = 0.14$, Table 3), in favor of LD-RT responders.

The 3 deaths (15%) observed at day 28 in the LD-RT cohort are described in Table 2. Two patients had COVID-19 symptoms escalate despite LD-RT; both went on to intubation and could not be weaned. The third was a patient with a large brain meningioma who died after family removed opted to remove supportive high-flow oxygen.

Contrary to published mortality rates, none of the 20 blindly selected controls died. No other toxicity, airway emergencies, or adverse events were observed following LD-RT.

Discussion

This report describes 28-day outcomes of an exploratory phase II trial of whole-lung, low-dose radiation therapy (LD-RT) given concurrently with standard drug treatments for COVID-19 related pneumonia.

Intubation rates declined 56% following LDRT compared to controls, from 32% to 14%, which approached significance despite a sample size of only 40 patients ($p = 0.09$, Fig. 2).

We report serologic reductions following LD-RT compared to controls: C-reactive protein ($p = 0.02$) and creatine kinase ($p < 0.01$), as well as cardiac marker troponin-1, liver function, and coagulopathies (Fig. 6), suggesting LD-RT may prevent long COVID sequelae and may be detectable with larger sample sizes.

Our study has numerous limitations: a blindly-matched but non-randomized design, exploratory intent, small patient numbers, differing control treatments, differing laboratory and imaging schedules between the LD-RT and control cohorts, variable times of symptom onset, limited imaging (Fig. 5) and serological (Fig. 6) studies in the control cohort before intervention and beyond 7 days, and lack of detailed viral load evaluations.

Conclusion

Patients whose CRP declined over 3 days following LD-RT had 100% survival without intubation, fewer prolonged hospitalizations, and lower burden of oxygen requirements.